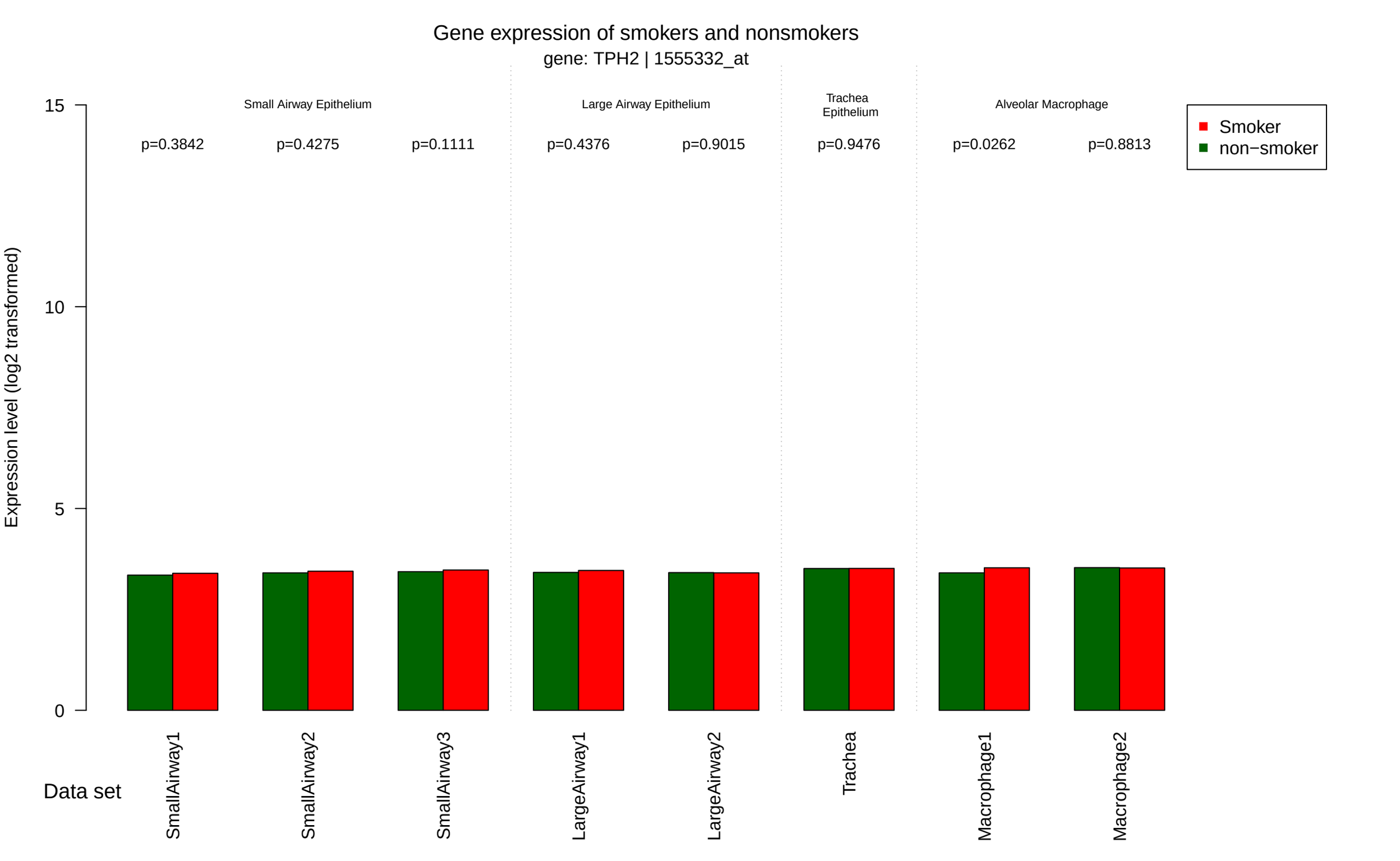
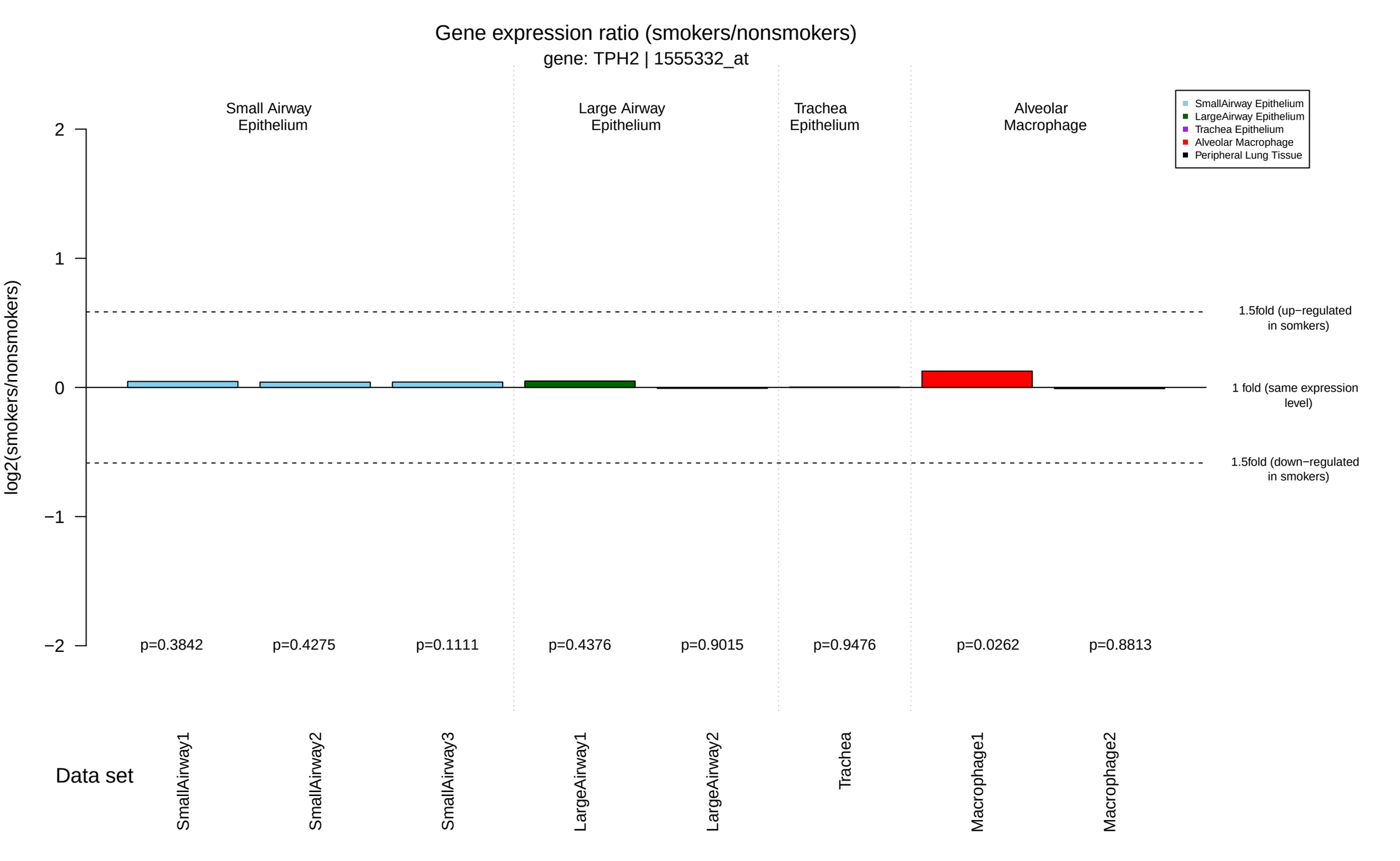
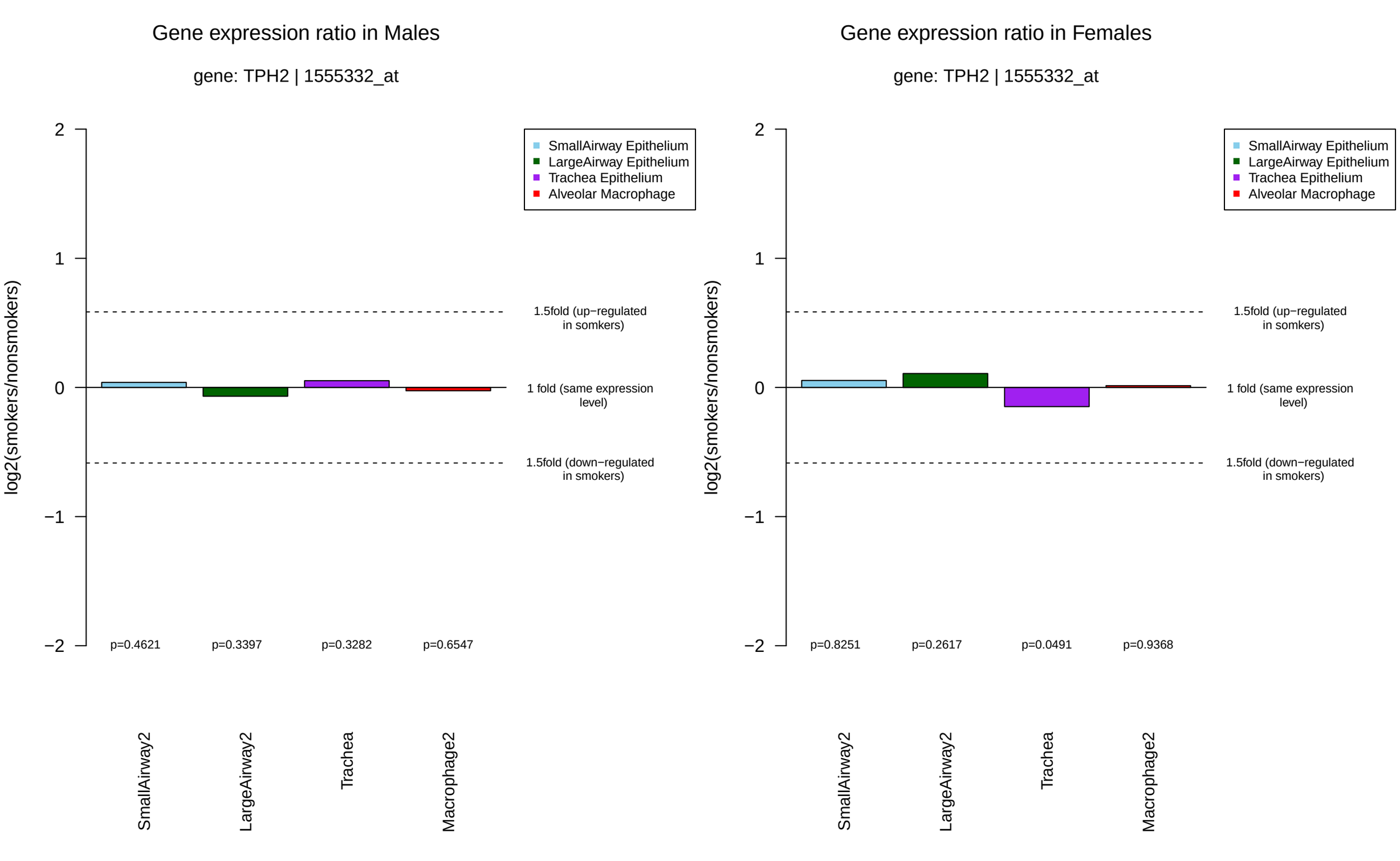
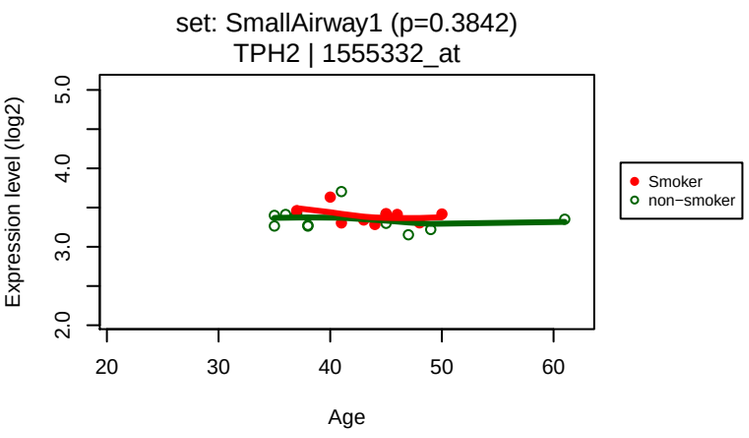
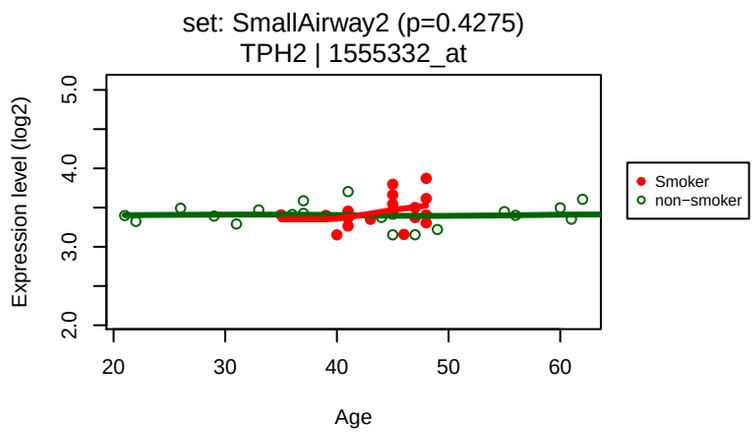
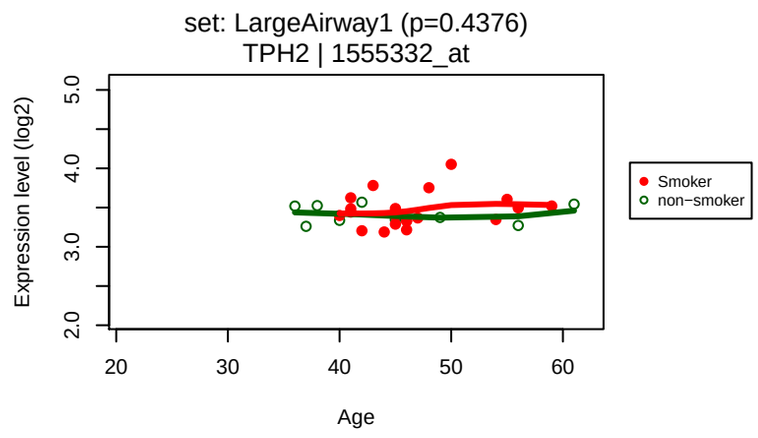

Gene expression of smokers and nonsmokers
gene: TPH2 | 1555332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3842
p=0.4275
p=0.1111
p=0.4376
p=0.9015
p=0.9476
p=0.0262
p=0.8813
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

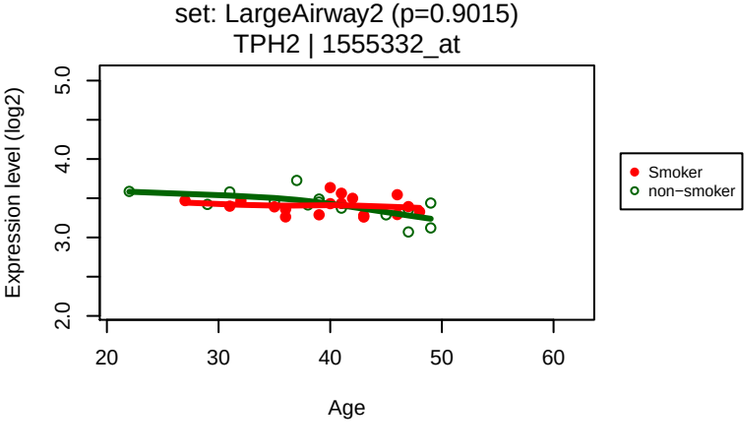

Gene expression ratio (smokers/nonsmokers)
gene: TPH2 | 1555332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3842
p=0.4275
p=0.1111
p=0.4376
p=0.9015
p=0.9476
p=0.0262
p=0.8813
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

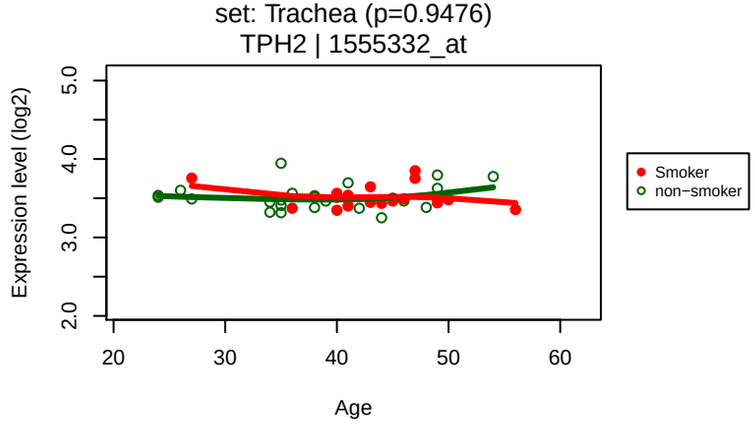

Gene expression ratio in Males
gene: TPH2 | 1555332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4621
p=0.3397
p=0.3282
p=0.6547
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TPH2 | 1555332_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8251
p=0.2617
p=0.0491
p=0.9368
SmallAirway2
LargeAirway2
Trachea
Macrophage2

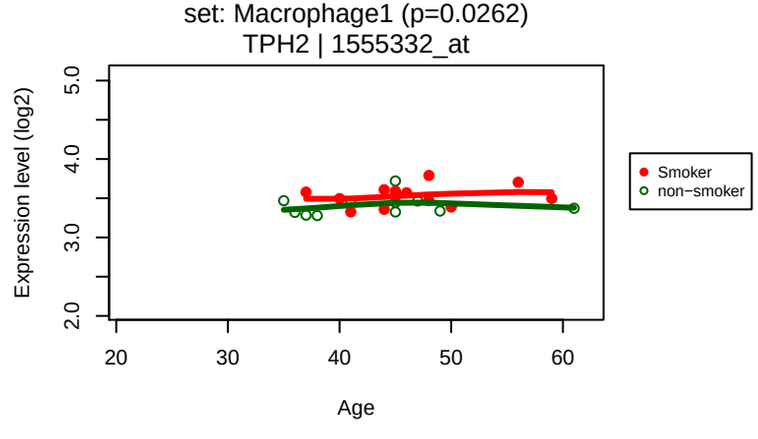

set: SmallAirway1 (p=0.3842)
TPH2 | 1555332_at
Expression level (log2)
Age
Smoker
non−smoker

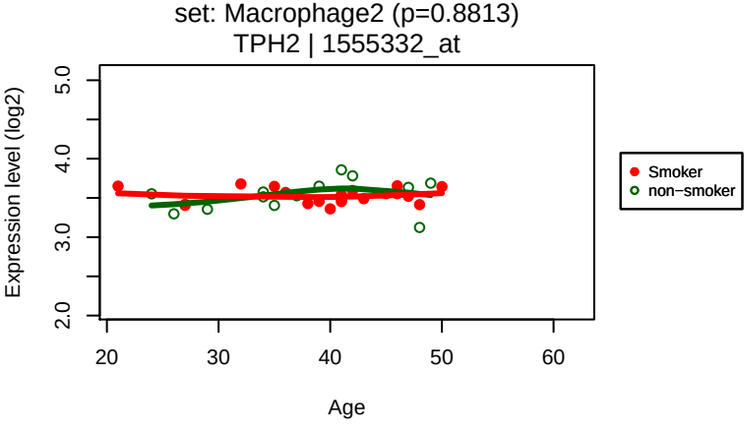

set: SmallAirway2 (p=0.4275)
TPH2 | 1555332_at
Expression level (log2)
Age
Smoker
non−smoker

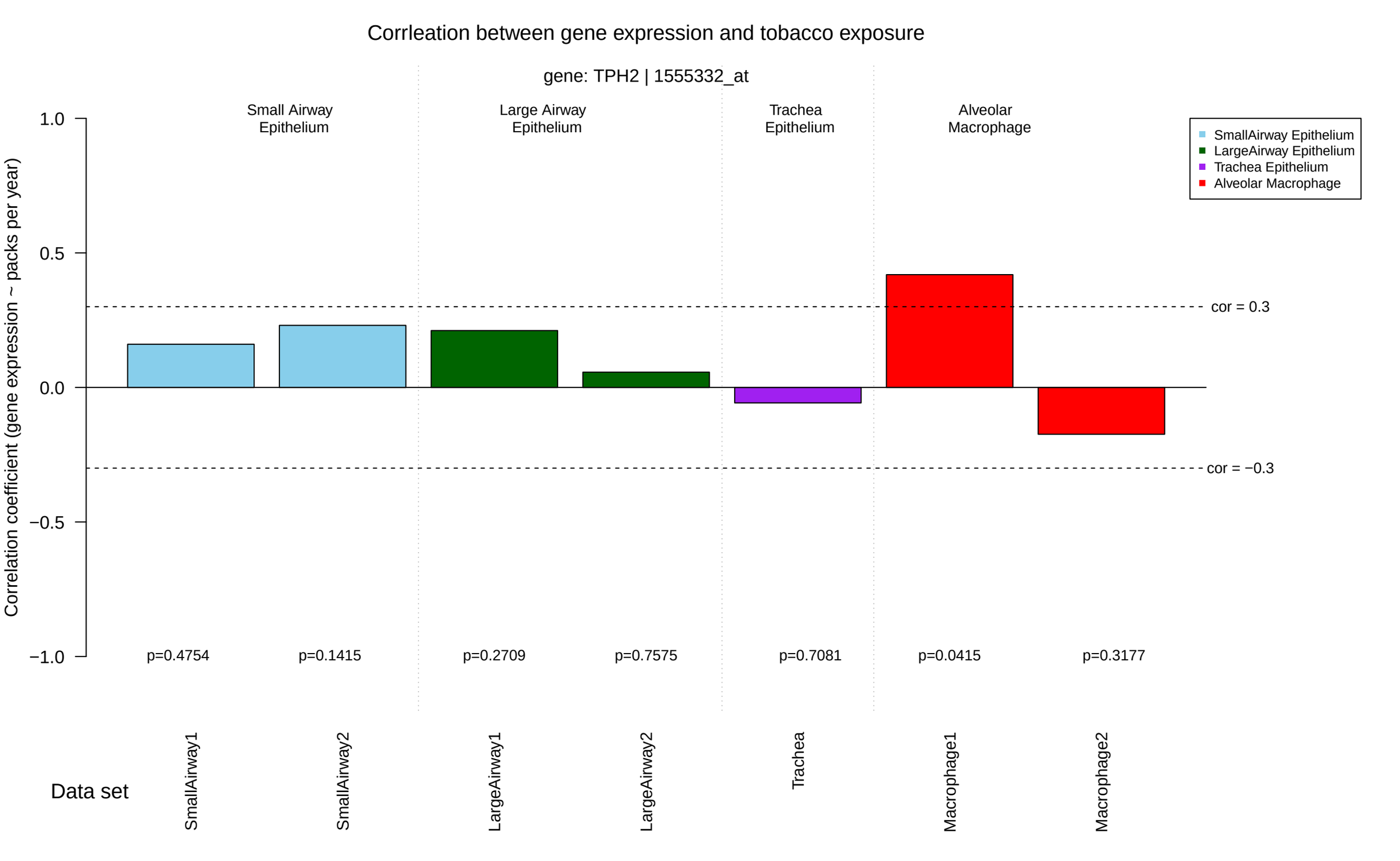

set: LargeAirway1 (p=0.4376)
TPH2 | 1555332_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9015)
TPH2 | 1555332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9476)
TPH2 | 1555332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0262)
TPH2 | 1555332_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8813)
TPH2 | 1555332_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TPH2 | 1555332_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4754
p=0.1415
p=0.2709
p=0.7575
p=0.7081
p=0.0415
p=0.3177
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2